TURP Syndrome *Prodigy Anesthesia*

Transurethral resection of the prostate (TURP) syndrome is a water intoxication syndrome that results from the absorption of irrigating fluid through open venous sinuses during transurethral prostate resection.

The average amount of fluid absorbed during a TURP is about 20 mL/min of resection time. The actual amount, however, depends upon the hydrostatic pressure of the irrigating fluid, the number and size of the venous sinuses exposed, and the duration of the resection.

The symptoms of TURP syndrome are related to the rapid increase in intravascular volume, the dilution of electrolytes (particularly sodium) and protein in the bloodstream, and the effects of the particular irrigation fluid used. The symptoms range from restlessness, nausea, dizziness, and shortness of breath to confusion, bradycardia, seizures, coma, and cardiovascular collapse. In the awake patient (regional anesthesia) the initial presentation is often an increase in blood pressure, increase in pulse pressure, bradycardia, and confusion. If glycine irrigation fluid is used, then the patient may exhibit encephalopathic symptoms related to high ammonia levels. Conversely, if sorbitol solutions are used, the patient may develop hyperglycemia, which can become severe in diabetic patients. Central nervous system symptoms of hyponatremia such as confusion, restlessness, nausea, and vomiting begin to appear at about 120 mEq/L with ECG changes occurring at 115 mEq/L and ventricular tachycardia and fibrillation occurring below 100 mEq/L.

Other complications of TURP include bleeding from the open venous sinuses (about 2-4 mL/min of resection time. Thromboplastin is found in high concentrations in cancerous prostate cells, and the procedure may result in the release of large quantities of it, resulting in a DIC-like coagulopathy, requiring the administration of coagulation factors and platelets. Perforation of the prostate capsule occurs in 1-2% of patients, resulting in the infusion of the irrigation fluid into the peritoneum. Symptoms may include hypertension or hypotension, poor return of irrigation fluid, and and abdominal pain, diaphoresis, and nausea in awake patients.

Nagelhout JJ, Plaus K. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 755-757.

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